

\$ seo-content-pipeline

The autonomous SEO operating system

It doesn't just recommend the fix — it ships the fix, measures the impact, and learns. A closed-loop SEO engine that lives in your stack.

Runs as a **Claude Code skill**

an **MCP server** (51 tools)

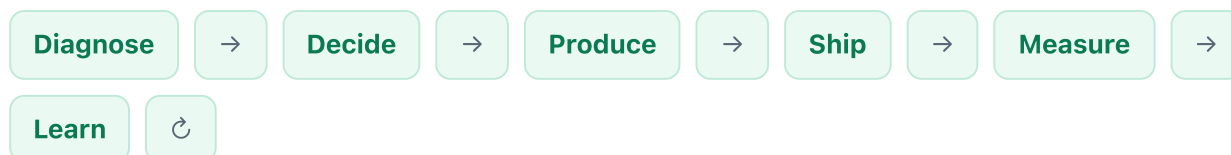
or a **standalone CLI**

Product brochure & quick-start manual · site-agnostic · file-based · every API optional

WHY IT'S DIFFERENT

99% of SEO work is diagnosis without execution

Audits, spreadsheets, and briefs that *someone else* has to implement. This tool collapses that gap. It lives where the work happens — the codebase and the data — and runs the whole loop end to end:



■ It ships the fix, not a description

Every finding becomes a real change — a git PR that edits meta/schema/redirects, a CMS update, or a browser action — gated by your approval.

■ It proves what worked

A causal change ledger logs every change and attributes the outcome — before→after clicks & position vs a **holdout of untouched pages**. No human does this rigorously.

■ Expert-grade by design

Outputs are written by named personas — a McKinsey-caliber strategist, a top technical SEO, an E-E-A-T content writer — not a generic model.

■ Built for the AI-search era

Tracks your citation share in ChatGPT / Perplexity / Gemini / Google AI Overviews, and optimizes entity + passage-citability to win it.

The moat: that closed loop — execution + change-level attribution — is the thing SaaS dashboards and human consultants structurally can't do. It compounds into a proprietary "what actually works on *this* site" dataset that sharpens every future decision.

One engine, the whole SEO lifecycle

■ Technical Doctor

JS-rendered crawl, Core Web Vitals, canonical/redirect/orphan/hreflang checks, sitemap & robots hygiene, structured-data generation + validation, log-file crawl-budget & AI-crawler coverage. Every finding ships a diff.

■ Strategy Brain

Topical-authority map, cannibalization → 301 plan, striking-distance keywords, internal-PageRank sculpting, competitor content & backlink gaps, and monthly competitor sitemap-deltas.

■ Content Studio

SERP-grounded briefs and full drafts, decaying-page refreshes, title/meta rewrites — written by a multi-agent crew (research → strategy → write → edit → tech-SEO).

■ AI-Search / GEO

AI-visibility tracker (mentions + citations + share-of-voice across ChatGPT/Perplexity/Gemini/AI Overviews), entity graph + Wikidata, passage-citability scoring.

■ Site Control

Full CRUD via repo PRs, WordPress/Webflow/Ghost APIs, or headless-browser automation — create, update meta/content, delete, redirect. Every change autonomy-gated and logged.

■ Reporting & Attribution

Daily/weekly/monthly digests, causal ledger with holdout attribution, GA4 organic revenue, decay radar, and anomaly alerts (indexation drops, CWV regressions, an AI Overview appearing).

Safe by design. A programmatic-content safety gate hard-blocks thin / near-duplicate / boilerplate pages before publish (Google's 2026 scaled-content enforcement). Nothing ships without passing the gate *and* your autonomy mode.

From zero to a shipping loop

1 Install

Python 3.11+. Core is numpy + scikit-learn; every API is optional.

```
# as a standalone CLI
$ pip install numpy scikit-learn
$ git clone <repo> && cd <repo>
# ...or drop it in as a Claude Code skill, or run: python -m seo_agent mcp
```

2 Create a workspace (one folder = one site)

```
$ python -m seo_agent init --site https://your-site.com
$ python -m seo_agent wizard # guided, hand-holding setup
```

The wizard walks you through every step with  /▶/○ status and the exact next action.

3 Wire the data (the readiness gate blocks until you do)

```
$ python -m seo_agent preflight # 0-100 readiness score
# Search performance: a GSC service account, or import an export:
$ python -m seo_agent gsc --csv export.zip
# Market data: DATAFORSEO_LOGIN / DATAFORSEO_PASSWORD in .env (or Semrush/Ahrefs)
```

Optional: PageSpeed key, GA4, Slack/WhatsApp channels, email transport — each unlocks more.
Nothing required to start.

4 Baseline & get the plan

```
$ python -m seo_agent onboard # → BASELINE.md (audit + speed + gaps + AI-search)
$ python -m seo_agent plan # ranked "what to do next"
$ python -m seo_agent consult # board-ready growth strategy
```

5 Choose how much it may do on its own

Set `autonomy`: **manual** (plan only) · **approve** (queue → you sign off) · **auto** (execute, with destructive changes still queued).

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Ship fixes & content — with human review

```
$ python -m seo_agent audit --fix          # audit + PR-ready remediation plan
$ python -m seo_agent pr fixes.json        # edit repo files → open a PR
$ python -m seo_agent crew article "keyword" # multi-agent draft → gated publish
$ python -m seo_agent review                # send to reviewers (Slack/WhatsApp/email)
$ python -m seo_agent apply --approved      # push ONLY approved items
```

Reviewers reply `APPROVE 3` or `CHANGES 3 make it punchier` from any channel. Only approved changes ship.

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Run on a cadence & deliver

```
$ python -m seo_agent run --daily --email    # pulse + auto-email the PDF
$ python -m seo_agent run --monthly          # full digest + AI-search + backlink gap
$ python -m seo_agent anomaly --alert        # regressions → your channels
```

Schedule via cron or the hosted scheduler. Reports go to email, Slack, or Mattermost.

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Measure & ask why

```
$ python -m seo_agent ledger                # every change → holdout-adjusted lift
$ python -m seo_agent explain /pk/lawn-suits # "why did this page drop?"
```

`explain` correlates a decline against our own change log, the GSC trend, and the Google-update timeline — evidence, not correlation theater.

Command map

Stage	Commands
Onboard & Doctor	<code>init</code> <code>wizard</code> <code>preflight</code> <code>onboard</code> <code>audit</code> <code>sitemap</code> <code>speed</code> <code>logs</code> <code>schema</code> <code>renderdiff</code> <code>llmstxt</code>
Observe	<code>ingest</code> <code>gsc</code> <code>gsc --csv</code> <code>rank</code> <code>trends</code> <code>backlinks</code> <code>ctr</code> <code>ga4</code>
Decide	<code>plan</code> <code>consult</code> <code>consolidate</code> <code>gap</code> <code>competitors</code> <code>aio</code> <code>pagerank</code> <code>decay</code> <code>algo</code> <code>authority</code> <code>eat</code>
Produce	<code>brief</code> <code>draft</code> <code>crew</code> <code>refresh</code> <code>retitle</code> <code>citability</code> <code>score</code>
AI-search	<code>aivis</code> <code>entity</code> <code>geo</code> <code>citability</code>
Control & review	<code>control</code> <code>pr</code> <code>webtask</code> <code>autonomy</code> <code>review</code> <code>approve</code> <code>changes</code> <code>apply --approved</code>
Measure & deliver	<code>ledger</code> <code>explain</code> <code>anomaly</code> <code>report --pdf --email</code> <code>run --daily --monthly</code> <code>email</code>
Scale (agency)	<code>projects</code> <code>jobs</code> <code>mcp</code>

51 tools, all also exposed over MCP. Everything degrades: with zero credentials you still get the full Site Doctor, strategy, drafts, entity/citability, and the plan.

HOW & WHERE IT RUNS

Your stack, your rules

■ Three runtimes, one engine

Claude Code skill · any MCP client / agent runtime · standalone CLI (cron). A hosted web app is planned — a thin wrapper over the same engine.

■ Human-in-the-loop by default

Nothing structural ships without review. Approve from CLI, email, Slack, Mattermost, or WhatsApp. Autonomy is a dial you control per site.

■ Your data stays yours

File-based, per-workspace. No central warehouse of Google/DataForSEO data — a compliance feature, not a limitation.

■ Fork-safe secrets

Keys live in a git-ignored, leak-scanned `.env`. They can never leak from a fork — enforced before anything else runs.

SEO Content Pipeline — product brochure & quick-start manual. Site-agnostic · file-based · 51 tools · runs as a Claude Code skill, an MCP server, or a standalone CLI. Every output is proposed and human-gated; content passes a programmatic-safety gate before publish.